8.2 Analyzing Arithmetic Sequences and Series

Name _____ Period ____ Date _____

Algebra II 8.2 (30-minute plan) · Student sheet

I can…

- tell whether a sequence is arithmetic and write its rule.
- write a rule from one term and the common difference, or from two terms.
- find the sum of an arithmetic series.

Vocabulary · fill in the missing word

Arithmetic sequence — the difference between _____ terms is constant

Common difference d — the _____ difference

Rule — a_n = _____

Arithmetic series — the sum of the terms of an arithmetic _____

Sum — S_n = _____

Part 1 · Notes and examples (fill in each note, then work the example)

Arithmetic sequence: Arithmetic: the _____ between terms is constant (d).

Example 1

Tell whether each sequence is arithmetic.

a. −9, −2, 5, 12, … b. 23, 15, 9, 5, …

Rule: Rule: a_n = _____

Example 2

Write a rule for the nth term of 4, 9, 14, 19, … Then find a_{20}.

Sum of an arithmetic series: Sum: S_n = _____

Example 3

Find Σ (i = 1 to 20) (3i + 7).

Part 2 · Partner practice

Partner practice

One term of an arithmetic sequence is $a_6 = 32$, and $d = -5$. Write a rule for the nth term.